

Utkarsh Patel

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SKILLS

Languages:	Python, SQL
ML & AI:	Pandas, NumPy, Scikit-learn, TensorFlow, Keras, XGBoost, LightGBM, joblib
Data & Analytics:	Tableau, Power BI, Matplotlib, Seaborn, Plotly, OLAP, PySpark
DevOps & Deployment:	Docker, Git/GitHub, Streamlit
Foundations:	Data Structures & Algorithms, DBMS, Normalizations
Soft Skills:	Collaborative, Problem Solving, Adaptability, Resilient, Diligent

PROJECTS

NYC Taxi Duration Estimator | Github

Jun' 2026

Python · Pandas · NumPy · Scikit-learn · XGBoost · LightGBM · pytest · joblib

- Analyzed 1.45M+ NYC taxi trips using Python, Pandas, and SQL-style validation; identified trip-duration skew, GPS anomalies, and temporal traffic patterns, then built a reproducible feature-engineering pipeline with Haversine distance, pickup-hour, day-of-week, and weekend features, removing 2.02% invalid records.
- Benchmarked Linear Regression, Random Forest, XGBoost, and LightGBM using RMSLE, RMSE, and MAE; tuned LightGBM achieved 0.364 RMSLE, improving predictive performance by 28% over the Linear Regression baseline.
- Performed feature-importance and residual analysis to identify distance and pickup hour as key predictors; evaluated blending and stacking and selected standalone LightGBM after stacking yielded <0.3% RMSLE improvement at 159× model size and 38× inference latency.

SQL Data Warehouse | Github

Aug' 2025

SQL (T-SQL) · OLAP · Tableau

- Architected an enterprise data warehouse with Medallion architecture (Bronze/Silver/Gold) layers for standardized CRM/ERP analytics.
- Automated ETL pipelines with SSIS, integrating CSV sources and applying cleaning and partitioning to drive 40% faster processing.
- Optimized star and snowflake schemas, developed 75+ SQL scripts across Medallion layers, and ensured 100% data integrity through validation and cleansing workflows.

Washington EV Population Analysis | Github

Jan' 2025

SQL · Python (NumPy, Pandas, Matplotlib, Seaborn, Plotly) · Tableau

- Analyzed Washington's EV population data (2000–2025) using SQL and Python, identifying 450% growth in EV adoption between 2010 and 2025.
- Processed and cleaned 150,000+ EV records using Pandas and NumPy for trend extraction across manufacturers, models, and counties.
- Built interactive Tableau dashboards with line, bar, and heat maps, and visualized growth patterns with Matplotlib, Seaborn, and Plotly, uncovering a 30% surge in registrations post-2020 due to state incentives.

CERTIFICATIONS

OCI Generative AI Professional | Oracle

Oct' 2025

OCI AI Foundations Associate | Oracle

Oct' 2025

ACHIEVEMENTS

Top 20 – AWS Hackathon

Feb' 2025

Secured 18th rank in “Prompt Builder 2025” organized by Student Organization Vibranta

EDUCATION

Lovely Professional University

Bachelor of Technology in Computer Science & Engineering | CGPA: 7.9/10

Phagwara, Punjab

Aug' 2023 – Present